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Trust, Confidence, and Expertise in a Judge–Advisor System

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The relationship between trust, confidence, and expertise in Judge–Advisor Systems is examined in two experiments with Judge–Advisor pairs, one with strangers and another with participants in ongoing relationships. There was expertise asymmetry so that Judges had less expertise than their Advisors. The dyads could receive money for accurate Judge decisions. Either the Judge or Advisor had the power to allocate this money between dyad members, before task interaction in study one and after task completion in study two. Because Judges were more dependent on Advisors than vice versa, it was predicted that trust would be more important to Judges. Results were supportive. Judges had higher and more variable ratings of trust in their partner than did Advisors, suggesting that Judges were more motivated to evaluate trust. High confidence by Advisors had a positive impact on Judges' ratings of trust and tendency to follow their advice. Judges' trust in their Advisors was significantly related their taking the advice and being confident in their final decisions. Although participants in study two had higher levels of trust in their partners, they allocated less money to them. The implications for establishing trust are discussed.

Decisions are often social in that decision makers use information provided to them by others. Often information is sought from someone perceived to know

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more or have a different perspective than one's own. People facing important decisions may solicit the advice of physicians, brokers, consultants, therapists, or even psychics. Uncertainty is aversive in most situations, and one way to reduce it is social information search, i.e., the solication of information from another person (Festinger, 1954; Schachter, 1959; Sniezek, 1995; Sniezek & Spurlock, 1996). To the extent that the decision maker can trust the advisor, uncertainty should be reduced even further. This study examines trust between a decision maker or Judge and an Advisor, in what is called a Judge–Advisor System (JAS).

The JAS paradigm (Sniezek, 1995; Sniezek & Buckley, 1995) avoids the extremes of studying only individual decision making or decision making by groups with identical roles for members. A JAS distinguishes between one or more persons who provide recommendations and information (Advisors) to the person with the responsibility for the decision (the Judge).

Judges typically depend on Advisors for information and opinions and possibly support for their decisions. The Advisor, in turn, may have some stake in the Judge's decision (Sniezek, 1995; Sniezek & Buckley, 1995). Outcome interdependency is not uncommon; the Advisor may get a percentage of the profits from the Judge's decision or suffer loss of reputation or job security following negative outcomes from the Judge's decision following advice. The studies in this article use a JAS with a Judge receiving information from one Advisor who has relevant expertise and a monetary stake in the Judge's decisions. Therefore, the Judge is dependent on the Advisor for recommendations and the Advisor is dependent on the Judge to make the best possible decision so that the dyad can receive money. The locus of reward power is manipulated so that either the Judge or Advisor has the power to allocate money between dyad members. The first case is based on an organizational model in which a decision maker can influence valued outcomes, e.g., monetary bonuses for another organizational member who assists the decision maker. The second case is based on a market model in which a consultant sets the fee for providing advice to clients.

TRUST, DEPENDENCE, AND UNCERTAINTY

When one depends on another person for information or opinions, one inevitably confronts "social uncertainty," an inability to predict perfectly the other person's behavior (cf. Messick, Allison, & Samuelson, 1987; Sniezek & Spurlock, 1996). One cannot predict all the motives and other influences on an Advisor's behavior. Often there is an information asymmetry between Judge and Advisors that fosters social uncertainty. A Judge may know little about a problem and consult an Advisor because of her expertise. Even with a detailed explanation, the Judge may not understand well enough to eliminate all uncertainty about whether the Advisor is giving the best possible information.

One factor that may reduce social uncertainty is trust. Trust is the expectation that the person is both competent and reliable and will keep your best interests in mind (Barber, 1983). Trust is also a "behavior that reflects a reliance on a partner and involves a vulnerability . . . on the part of the trustor" (Moorman, Zaltman, & Deshpande, 1992). For there a Judge to trust an Advisor, there must be some risk of getting advice that is not good for the Judge. Trust cannot emerge without social uncertainty (Kollock, 1994). If one can be certain about another's behavior because one has perfect information about it or the other person's behavior is forced, there is no social uncertainty or risk, and trust is irrelevant. Trust reduces social uncertainty by limiting the range of behavior expected from another (Kollock, 1994).

Kollock (1994) simulated two types of markets differing in social uncertainty. One market had no social uncertainty; sellers had to reveal the exact quality of their goods to the buyer before the exchange. Another market had high social uncertainty; buyers did not find out the quality of goods until after they bought them. This market left opportunity for deception due to the information asymmetry between the buyers and sellers. Kollock found trust was more important to participants when there was high social uncertainty. In this condition, participants were more committed to one specific partner and developed more trust in their partners. They also had more variability in their trust ratings, using more extreme ends of the trust scale and rating partners as either very trustworthy or very untrustworthy.

There are three ways trust can be examined (Dwyer & Lagace, 1986; Lewis & Weigert, 1985), e.g., as the personality trait of propensity to trust (Barber, 1983; Mayer, Davis, & Schoorman, 1995; Rotter, 1967, 1980) or as a generalized expectancy about the behavior of another person (Johnson-George & Swap, 1982). Third, trust can be examined as a behavioral manifestation (Mayer et al., 1995); for example, as cooperation in a prisoner's dilemma (Deutsch, 1962; Dwyer & Lagace, 1986). Schlenker, Helm, and Tedeschi (1973) defined trust as a behavior, "trusting behavior entails an action that manifests one's evaluation that the risk of becoming vulnerable or dependent on another is worth the possibility of a positive outcome. Trust is reliance upon information received from another person in the face of such risk." The last two conceptions of trust are relevant here. The present study examines trust as the Judge's behavior of relying on and using information provided by the Advisor during the decision task. In addition, it examines trust as an expectancy about one's partner's behavior via a posttask questionnaire.

CONFIDENCE

Another variable that is important in reducing uncertainty about advice is the level of confidence of the Advisor. Confidence is the strength with which a person believes that a specific statement, opinion, or decision is the best possible (Peterson & Pitz, 1988; Zarnoth & Sniezek, 1999). High Advisor confidence can act as a cue to expertise and therefore influence the Judge to accept the advice (Sniezek & Buckley, 1995). An important question is whether Advisor confidence is a valid cue to advice quality, i.e., whether confidence assessments predict accuracy, either over Advisors (interindividual validity) or over multiple tasks for one Advisor (intraindividual validity).

STUDY OVERVIEW

Participants were assigned to roles based on their expertise regarding computer knowledge and operation: Judges had low expertise, and Advisors had high. Judges were paired with an Advisor to work on a multiple-choice computer knowledge task in which participants could win money based on the correctness of the Judge's final answers.

Advisors provided their recommended choices and accompanying confidence assessments to the Judge and, if desired, elaborated with comments as well. Judges were cued, not independent (Sniezek & Buckley, 1993, 1995; Sniezek, Paese, & Switzer, 1990). That is, they were given the question and the recommendation simultaneously without previously making their own independent (tentative) choices. Cueing is common with information asymmetry because Judges with low expertise have difficulty making an independent choice. Compared to independent choice, cueing tends to reduce information search and accuracy and increase confidence (Sniezek & Buckley, 1995; Sniezek et al., 1990). The Judge gave a final decision and confidence assessment for each item.

Participants interacted with their partner through writing in both experiments. Participants did not see or know their partners in the first experiment. In contrast, they interacted face-to-face with peers with whom they often had ongoing relationships in the second.

In both experiments, reward power was manipulated with one condition exemplifying the market model and another exemplifying the organizational model. That is, either the Advisor had the reward power to determine what percentage of money to charge for their advice (Advisor-power) or the Judge had the reward power to decide what percentage of the money to pay the Advisor (Judge-power). In experiment one, a basis for initial trust of Judges by Advisors and vice versa was provided by having the person without reward power learn the allocated amount before the task, which could reduce social uncertainty about the other partner. The partner with reward power was not only deciding but also informing his partner of his risk of being unfairly compensated. For example, a Judge with reward power could reduce his Advisor's fear that he was working hard at advising only to have the Judge take all the money. In addition, an Advisor with reward power could be relieved by the expectation that his Judge would find the allocation acceptable and thus be less likely to ignore the advice and guess on the task. Of course, the same logic concerning this initial basis for developing trust—or mistrust should allocations to the partner be trivial—applies equally to Judges and Advisors, so it will not contribute to differential trust between the two roles. Although the percentage allocated could also influence later Judge–Advisor interactions in an interesting manner, sample sizes and low variability in the allocations here preclude formal analysis of such influence.

In experiment two, allocation decisions were made at the very end of the experiment, and were not revealed to the partner without power, thus they cannot play a role in the development of trust. In this case, the percentage allocated was an outcome of the interaction. When participants know one another, there is less need to reduce social uncertainty at the start of their exchange, thus reward decisions may follow the advice. In summary, the primary differences between the two experiments were the nature of the Judge and Advisor's prior relationship and interaction during advising and the timing and communication of the monetary allocation decision.

HYPOTHESES

Figure 1 illustrates the model from which the hypotheses are derived.

Uncertainty and Trust

Trust should be more important to the partner with less expertise, the Judge. The Judge has more uncertainty about the task and therefore will be more dependent on the Advisor and have more need to develop trust than does the Advisor.

Hypothesis 1a: Judges should develop higher levels of trust in their Advisors than their Advisors do in them and therefore give higher ratings of trust than do Advisors.

Hypothesis 1b: Because Judges need to decide whether to depend on an Advisor for advice, they should use more discrimination in forming trust in their partner and have more variability in their trust ratings than the Advisor. In other words, Judges should be less likely than Advisors to have neutral levels of trust in their partner. This prediction follows from the results of Kollock's (1994) study in which participants' with uncertainty had more variance in trust ratings.

Hypothesis 1c: As the expertise asymmetry, as measured by the computer pretest, increases between Judge and Advisor, Judges should become more dependent on the Advisor and develop higher levels of trust.

Cues to Expertise

Hypothesis 2a—Confidence: Advisor confidence should act as a cue to Advisor expertise and be positively related to Judge trust.

Hypothesis 2b—Advice elaboration: Advisors' comments elaborating on their advice by giving Judges information or expressing confidence should be positively related to Judges' trust.

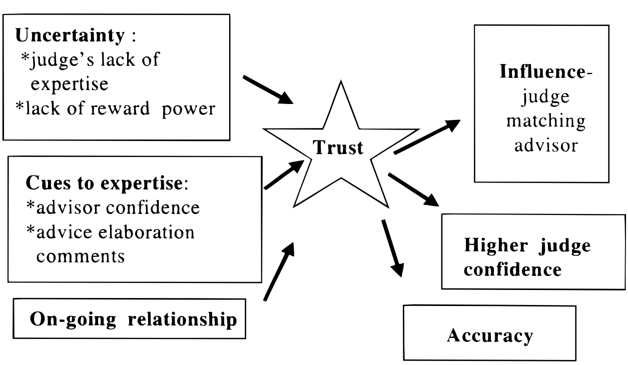

FIG. 1. Model of trust.

Hypothesis 2c—Influence of confidence: Higher Advisor confidence should influence the Judge, increasing the extent to which the Judge matches the Advisor's recommendations

Effects of an On-Going Relationship

Hypothesis 3: In experiment two, knowing or previously interacting with one's Advisor should be positively related to the Judges' trust.

Outcomes of Trust

Hypothesis 4a: Matching Advisors' answers should be positively related to Judges' trust.
Hypothesis 4b: Higher Judge confidence should be positively related to Judges' trust
Hypothesis 4c: High Judge accuracy should be positively related to Judges' trust.

EXPERIMENT 1

Method

Participants Participants were 152 undergraduates who completed the experiment for course credit. Twenty-five percent of dyads who scored in the top half were randomly selected to receive money: 50¢ for each correct answer in the 40 items up to a maximum of $20. This was divided between the Judge and Advisor according to the allocation decisions of the partner with reward power and paid a few weeks after the experiment.

Design. The design was 2 × 2 with the independent variables of reward power (Judge-power vs Advisor-power) and role (Judge or Advisor). Figure 2 shows the procedure for assigning participants to experimental conditions. Role condition was determined by their pretest score and assignment to the reward power condition was random.

Task. The task included 40 multiple-choice items on computer knowledge and operations. Prior to the task, participants took an 18 item pretest with similar questions. (Example: "This file contains characters that are in machine readable form. A. DOS file B. hidden file *C. binary file D. root directory file.")

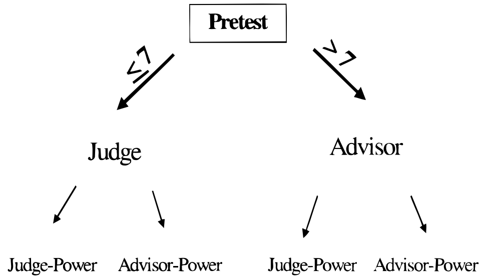

FIG. 2. Assignment of participants to levels of the independent variables.

A pilot study found the pretest to predict performance on the 40 items, $r = .67$, $p < .0001$. A posttask questionnaire was developed to measure trust and effort in task as well as prior computer experience.

Procedure. Participants read and signed informed consent forms explaining the study and the role of money. They were told the pretest would determine their role in the experiment. Participants took the 18 item pretest automated on computers and were given their role assignments: those who scored above 7 became Advisors and participants who scored 7 (the median performance in the pilot data) or below became Judges. This procedure does not perfectly distinguish between experts and nonexperts due to some chance element in the scores. This was intended to reflect the reality that distributions of experts and nonexpert often overlap over some fixed criteria like a cutoff for a board exam or a sales quota.

After the pretest, Advisors moved to another room and dyads were formed and assigned to conditions. Advisors and Judges did not know the identities of partners, but the nature of the procedure made it salient that they were interacting with another student in real time. Judges and Advisors communicated through written forms delivered by an experimenter.

After assignment to conditions, participants received a sheet explaining their role and the study. They were informed that the next part of the study was a 40-item computer test and told whether they were a Judge or Advisor and what this entailed. They were informed that Advisors scored in the top half while Judges had not, but not given their actual score (to minimize differences among the Judges and among the Advisors about perceptions of their abilities). The sheet explained how they could receive money and they were told which partner had the power to allocate money. Finally, the sheet asked each participant to either accept or decline to be in the experiment. The option to decline was used to increase realism and ensure that participants wanted to give or receive advice and were comfortable allocating money to another person. If they accepted, they had to allocate a percentage of the money to their partner if they had the power to do so. Those who declined ($n = 28$) worked on the 40 computer items individually. Participants who accepted were given an instruction sheet, and those without reward power received a form listing the percentage of the money that had been allocated to each partner.

At this point, Advisors were given the first set of 10 questions, each of which asked for a recommendation and confidence assessment (subjective probability of correctness) from 0 to 100. Advisors were instructed that they had the option of writing comments to the Judge on the advice form. Judges waited (3 to 7 min) for their Advisors' first set of recommendations to be completed and brought by the experimenter. Judges read the advice and entered their final decisions and confidence assessments for these items into the computer. This procedure was repeated for three additional sets of 10 questions each. Then participants completed posttask questionnaires and were debriefed.

Results and Discussion

Manipulation checks As was true in the pilot data, Advisors' (n = 60) scores on the pretest and performance on the 40 items were correlated, $r = .70$, $p < .0001$. A 2 (Judge vs Advisor) × 2 (Advisor-power vs Judge-power) ANOVA performed on the pretest scores verified that Judges' scores (M = 5.17, SD = 1.40) were significantly lower than Advisors' (M = 10.20, SD = 2.33), $F(1, 116) = 198.79$, $MS_E = 3.76$, $p < .0001$. Further, the samples of Judges and Advisors differed in that Advisors (M = 1.90, SD = 1.95) took significantly more computer classes than Judges (M = 1.03, SD = 1.19), $t(98) = 2.91$, $p < .005$. Advisors (M = 71.67%, SD = .45) were more likely to own a computer than Judges (M = 55.93%, SD = .50), $t(117) = 1.80$, $p < .10$. Estimates of the percentage of students knowing less about computers than oneself were higher for Advisors (M = 52.07%, SD = 22.03) than Judges (M = 28.17%, SD = 16.83), $t(105) = 6.43$, $p < .0001$. There were no interactions with locus of reward power.

Uncertainty and trust. Ratings for the eight posttask items relating to trust in partner were highly intercorrelated with a Cronbach's α of .90 for Judges and .78 for Advisors. Judge and Advisor trust consisted of the composite of their responses on these eight items. A MANOVA showed a significant difference in mean level of trust between Judges and Advisors, Wilks' $\lambda = .84$, $F(8, 104) = 2.55$, $p < .01$. In support of hypothesis 1a, Judges were generally higher than Advisors (see Fig. 3 and Table 1 for questions and means).

Next we analyzed participants' variability in trust ratings. First, scales for the items were "folded" at the midpoint and recoded so that the values 1 *(Strongly Disagree)* through 9 *(Strongly Agree)* became 5, 4, 3, 2, 1, 2, 3, 4, and 5, respectively. This permitted testing whether participants differed in their use of extreme ends of the scale, regardless of direction. A 2 (reward-power) × 2 (role) MANOVA conducted on the eight recoded items revealed a significant main effect of role, Wilks' $\lambda = .72$, $F(8, 104) = 5.11$, $p < .0001$. Judges used more extreme ends of the scale than Advisors, supporting hypothesis 1b (see Fig. 4).

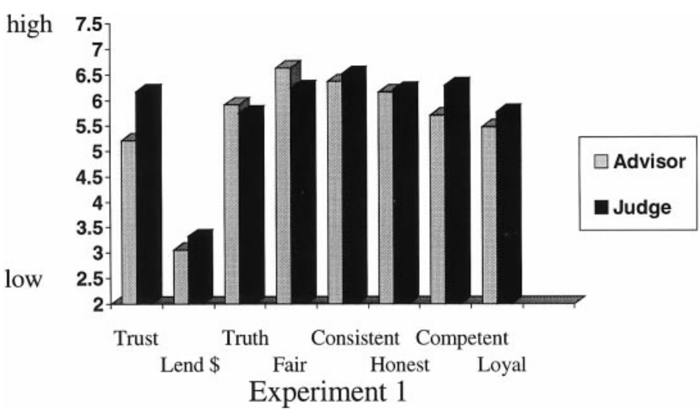

FIG. 3. Judge and Advisor means graphed for each of the eight posttask questionnaire items measuring trust (see Table 1 for questions) for Experiment 1.

TABLE 1

Judge and Advisor Means on Questions Measuring Trust in Experiment 1

Question	Judge		Advisor	
	M	*SD*	*M*	*SD*
Compared to other students in your class, how much did you *trust* your Judge/Advisor?*	6.17	1.62	5.21	1.06
I would be willing to *lend* my Judge/Advisor almost any amount of *Money* because he/she would pay me back as soon as he/she could.	3.32	2.19	3.05	1.66
I could expect my Judge/Advisor to tell me the *truth*.	5.75	1.92	5.91	1.93
I would expect him/her to play *fair.*	6.24	2.03	6.63	1.78
My Judge/Advisor *consistently* gave his/her best answer/advice.	6.51	1.76	6.36	1.46
My Judge/Advisor was *honest* in this task.	6.20	1.87	6.16	1.35
My Judge/Advisor was *competent* to serve as a Judge/Advisor in this task.*	6.29	1.79	5.70	1.25
My Judge/Advisor was *loyal* to me and had motives that were in my best interest.	5.76	1.97	5.48	1.26

Note. Questions were in a 9-point Likert-type format with a higher score indicating a higher rating of trust.

* Significant difference between the means at $p < .01$.

Hypothesis 1c predicted that Judge trust would increase as the expertise asymmetry between the Judge and Advisor became greater. To test this, the discrepancy between pretest scores of Judge–Advisor pairs was correlated with Judge trust. Hypothesis 1c was supported by a significant relationship ($n = 60$), $r = .25$, $p < .05$.

Predictions regarding the consequences of uncertainty that arises from a lack of expertise were supported. Compared to Advisors, Judges, who had more uncertainty in terms of expertise, developed higher levels of trust and gave more discriminating trust ratings.

Cues to expertise: Advisor confidence. The next set of hypotheses concerned two cues to expertise, Advisor confidence and advice elaboration, and their

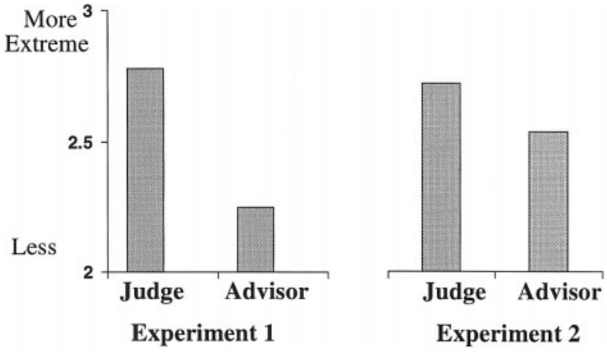

FIG. 4. Judge and Advisor means on use of extreme ends of the trust scale for Experiments 1 and 2.

relation to trust as expressed by either a rating or matching behavior. Hypothesis 2a, which predicted that Advisor confidence would be positively related to Judge trust, was supported by a significant correlation between Judge trust (n = 59) and Advisor mean confidence (r = .35, $p < .01$). Further analysis showed Advisor confidence to be a good cue to Advisor accuracy, with an interindividual validity of r = .71, $p < .0001$. In addition, Advisor confidence assessments had intraindividual validity, with correlations between r (number correct) and p_i for individual advisors ranging from −.21 to .58, with an average (computed using an r-to-Z transformation) of r = .26. Of these validity coefficients, 43% were positive and significant.

Advice elaboration. Hypothesis 2b concerned the relation of advice elaboration—which could serve as an additional cue to expertise—to Judge trust. Comments were given on 20% of the items and split into three categories. Confidence comments, such as “I’m definite about this” or “This is a guess” were the most frequent, occurring on an average of 9.6% of 40 items. Informative comments such as “Gigo means garbage in garbage out” or “Elm and Pine are mail systems” were made on 8.5% of questions. Advisors’ comments on their expertise such as, “Trust me, I know a lot about computers” appeared on only 2% of questions and thus were excluded from further analysis. In support of hypothesis 2b, the correlation between Judge trust and comment frequency was significant for both the information (r = .35, $p < .01$) and confidence categories (r = .28, $p < .05$).

Matching. The final prediction regarding cues to expertise, hypothesis 2c, stated that higher expressed confidence by Advisors would influence Judge matching, i.e., using the Advisor’s recommendation as the final decision. To test this, we compared Advisors’ confidence level when the Judges did and did not match. The difference of 18.40 was significantly different from 0, t = 8.41, $p < .0001$. To ascertain the wisdom of Judges’ matching decisions, we did a similar comparison of Advisors’ accuracy rates when matched and when not matched and found the difference of 19.85% significantly different from 0, t = 6.06, $p < .0001$. Thus, Judges were more likely to follow advice when it was correct, which is to be expected if they are using Advisor confidence as a cue to expertise and the cue is valid. Thus, Advisor confidence assessments are an important means of influence on the Judge.

Overall, Judges matched the Advisors’ recommendation on 80.79% of the items. However, amount of matching varied by reward power, with less in the Advisor-power condition (M = 75.37%, SD = .20) than in the Judge-power condition (M = 87.88%, SD = .11), $t(52)$ = 3.08, $p < .005$. Results discussed later offer a possible explanation for this difference.

Outcomes of trust. Next, data were analyzed to test hypotheses 4a, 4b, and 4c about the outcomes of Judge trust, specifically judge matching, confidence, and accuracy, respectively. Each hypothesis was supported: over Judges, Judge trust (n =59) was positively correlated with the amount of matching, r = .29, $p < .05$, Judge confidence (mean p_i) (r = .40, $p < .005$), and Judge accuracy

($r = .26$, $p < .05$). These outcomes of trust were beneficial because, in support of hypothesis 4c, trust was positively related to accuracy.

Money allocation. There was no difference between the Advisor-power ($M = 47.21$, $SD = 15.03$) and Judge-power ($M = 45.38$, $SD = 11.99$) conditions in the percentage given to the Advisor, $t = .51$ (58), $p > .10$. Most participants followed equality norms and split the money 50/50; thus, the basis for initial trust was common for all participants. The lack of variability in the allocations prevents analysis of them as a form of influence.

Posttask questionnaire. A role by reward power ANOVA conducted for the question "What percent of your maximum available effort did you spend trying to answer the items correctly?" showed a significant interaction, $F(1, 115) = 9.66$, $MS_E = 446.18$, $p < .005$. In the Advisor-power condition the mean response was 73.48% ($SD = 24.09$) for Judges and 78.18% ($SD = 22.11$) Advisors. However, in the Judge-power condition, Judges reported putting in 59.69% ($SD = 22.56$) and the Advisors 88.65% ($SD = 12.45$). Perhaps when Judges pay the Advisor, they do not feel they should work. This reported lack of effort is backed up by the matching data (see previous results). Judges in the Judge-power condition matched Advisors' responses significantly more often and reported putting forth less effort than Judges in the Advisor-power condition.

Conclusions

Results highlight the importance of trust in Judge Advisor Systems composed of strangers with minimal interaction. All hypotheses concerning expertise asymmetry were supported. Participants with less expertise (Judges) had higher trust and more variability in their trust ratings. The uncertainty associated with a lack of expertise is important for the evaluation and development of trust. In fact, the less expertise Judges had, the more they trusted the Advisors. These findings support Kollock's (1994) study showing the important of social uncertainty to the development of trust in markets.

Having demonstrated that Judges did develop trust, the next step is to explain how this occurs. Initial social uncertainty is a prerequisite for the development of trust as this uncertainty is reduced. In this research, high Advisor confidence, and Advisor comments giving information or expressing high confidence, were positively related to Judges' trust, and the former cue was related to Judge matching behavior. These results suggest that trust is important to the acceptance of advice and that Advisor confidence, Advisor comments, uncertainty, and dependence on another person are important to the establishment of trust.

Finally, all proposed outcomes of Judge's trust were supported. Judge trust was positively related to matching the Advisor, having confidence in that advice, and having correct answers. Once Judges develop trust in their Advisors, they are more likely to follow the Advisor's recommendations and be confident in their final decision. Due to the high validity of Advisors' expressions of confidence, the Judges who developed higher trust were more likely to be correct.

These results display the importance of trust and confidence in taking advice. However, often when we seek advice, we know or at least able to see our Advisor. In this experiment, participants could not see and did not know their partner. Thus, the results may not generalize to settings with more social interaction between Judge and Advisor either before or during the decision task. While seeking advice from a stranger is common in the market model, in the organizational model, a Judge would usually have some social information about the Advisor. To improve generalizability, we conducted experiment two using participants with ongoing relationships. The social cues available with an ongoing relationship are expected to provide a means of reducing social uncertainty and therefore be related to trust.

EXPERIMENT 2

Method

The methods and procedures were essentially the same as those for experiment one. In this section, we note only those aspects that were different.

Participants. Participants were 110 undergraduates in an upper level organizational psychology course who completed the experiment as a class exercise and volunteered their data for research.

Procedure. Participants completed the experiment together in class several days after taking the pretest. At the start participants received a folder with either the Judge or Advisor instructions and materials, depending on their role assignments, and a seat number. Pairs sat with one empty seat between partners. Unlike the first study, each participant could see his or her partner as well as the other Judge–Advisor dyads throughout the experiment.

Participants with the power to allocate money were instructed to allocate a percentage of the money at the *end* of the experiment, as can occur in the organizational model. Their partners were never informed of this allocation percentage. Any money received by the partner without reward power was the product of this allocation percentage and the number of correct Judge's answers—neither of which was revealed to them.

Advisors completed the first set of items in blue ink, then handed the form directly to their Judge. There was no oral communication. Judges made their final decisions and confidence assessments for this first set of items on the same sheet in black ink. Participants repeated this procedure for the remaining three sets of items. Then the participant with reward power decided on an allocation percentage, and both participants filled out the posttask questionnaire.

Results and Discussion

Manipulation checks. Advisors' pretest scores (n = 50) correlated with the their scores on the 40 items, r = .59, $p < .0001$. Judge and Advisor pretest

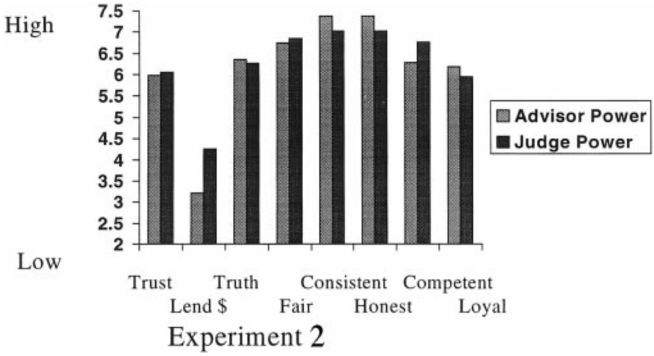

FIG. 5. Judge-power and Advisor-power means graphed for each of the eight posttask questionnaire items measuring trust (see Table 2 for questions) for Experiment 2.

means were similar to experiment one. A role × reward power ANOVA showed Judges' scores (M = 5.48, SD = 1.46) were lower than Advisors' scores (M = 10.08, SD = 1.89), $F(1, 96) = 186.12$, $MS_E = 2.86$, $p < .0001$. As in experiment one, Advisors had significantly more computer experience than Judges.

Uncertainty and trust. Ratings on the trust items were highly intercorrelated with a Cronbach's coefficient α of .87 for both Judges and Advisors. A MANOVA found a significant condition effect, Wilks's $\lambda = .85$, $F(8, 98) = 2.18$, $p < .05$. The Judge-power condition had higher trust than the Advisor-power condition (see Fig. 5 for a graph). Although Judges had higher trust than Advisors, this was only marginally significant, Wilks' $\lambda = .86$, $F(8, 98) = 1.96$, $p = .06$ (see Table 2 for means and Fig. 6 for a graph), thus failing to support

TABLE 2

Judge and Advisor Means on Questions Measuring Trust in Experiment 1

	Judge		Advisor	
Question	*M*	*SD*	*M*	*SD*
Compared to other students in your class, how much did your *trust* your Judge/Advisor?*	6.36	1.43	5.67	1.26
I would be willing to *lend* my Judge/Advisor almost any amount of *Money* because he/she would pay me back as soon as he/she could.	3.98	2.40	3.50	2.06
I could expect my Judge/Advisor to tell me the *truth*.	6.42	1.73	6.20	1.51
I would expect him/her to play *fair*.	6.82	1.98	6.78	1.44
My Judge/Advisor *consistently* gave his/her best answer/advice.	7.13	1.58	7.26	1.43
My Judge/Advisor was *honest* in this task.	7.29	1.47	7.09	1.34
My Judge/Advisor was *competent* to serve as a Judge/Advisor in this task	6.73	1.57	6.31	1.98
My Judge/Advisor was *loyal* to me and had motives that were in my best interest.	6.33	1.55	5.78	1.46

Note. Questions were in a 9-point Likert-type format with a higher score indicating a higher rating of trust.

* Significant difference between the means at the $p = .01$.

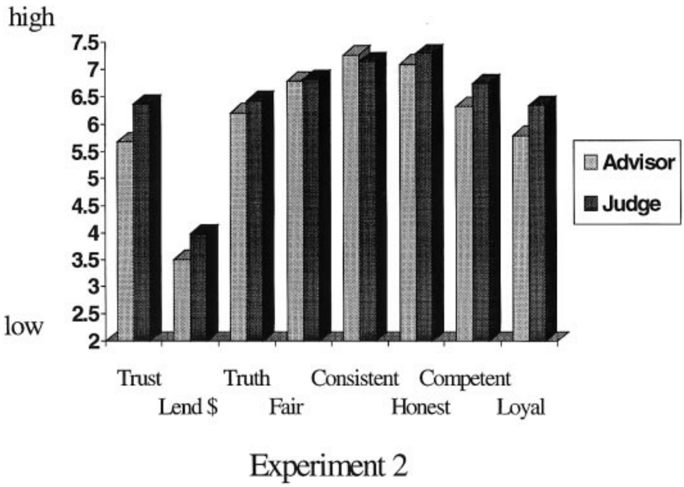

FIG. 6. Judge and Advisor means graphed for each of the eight posttask questionnaire items measuring trust (see Table 2 for questions) for Experiment 2.

hypothesis 1a. Knowing and seeing partners may have influenced trust ratings, thereby diluting the effect of the role manipulation.

Variability in trust ratings was examined in a 2 (reward power) × 2 (role) MANOVA on the recoded trust items. In support of hypothesis 1b, Judges gave more extreme ratings than Advisors, Wilks' $\lambda = .85$, $F(8, 98) = 2.14$, $p < .05$ (see Fig. 4 for a graph). Hypothesis 1c, which predicted higher Judge trust with increasing expertise asymmetry, was not supported.

Cues to expertise. Advisors' expressions of confidence showed high validity (interindividual $r = .74$, $p < .0001$, intraindividual average $r = .27$ with range $r = .01$ to .54 and 40% of r's positive and significant). Nevertheless, neither this cue to expertise nor the advice elaboration cue appeared to influence Judge trust. Judge trust ($n = 59$) was not significantly correlated with Advisor mean confidence or frequency of comments, failing to support hypotheses 2a and 2b.

Matching. Judges matched the Advisors' answers on 80.77% of the questions. When they did match, the Advisor's confidence was an average of 19.4 points higher than when they did not match—an amount which was significantly different from 0 and supportive of hypothesis 2c, $t = 8.06$, $p < .0001$. Similarly, the difference in accuracy rates between matching and not matching, 30.29%, was significantly different from 0, $t = 7.92$, $p < .0001$. These results suggest that high Advisor confidence influenced the Judge to match the Advisor's response. Because Advisor confidence was valid, Judges were more likely to get items correct when they matched the Advisor.

Thus far, the primary differences from experiment one are the findings pertaining to the development of trust as measured by ratings. When participants could see each other and already knew each other before starting the experiment, the dependence following from role and expertise asymmetry did not play the same role in explaining Judge's trust. Further the cues to expertise that were available during the experiment itself—Advisor confidence and advice elaboration—no longer relate to the Judges' ratings of trust. Here trust could be related more to liking or previous interaction with the person. In experiment one, trust may have been based more on perceptions of ability and expertise, while in experiment two, trust may have been based more on observable characteristics or previous knowledge of the person. This interpretation is supported by a positive correlation between Judge trust and whether the Judge knew something of the Advisor's background (r = 28, $p < .05$), had spoken to the Advisor in class about matters of work ($r = .28$, $p < .05$), or had interacted with the Advisor outside the classroom on matters of work ($r = .27$, $p < .05$). This supports hypothesis 3.

Outcomes of trust. Judge trust ($n = 55$) was correlated with the amount that the Judge matched the Advisor ($r = .32$, $p < .05$), Judge mean confidence ($r = .29$, $p < .05$), but not Judge accuracy ($r = .23$, $p > .05$). These results support hypotheses 4a and 4b, but not hypothesis 4c.

The relationship between uncertainty and trust is less supported in experiment two, and only two of the hypothesized outcomes of Judge trust are significantly related to trust. Judge trust also is not related to Advisor cues to expertise given during the task, but is related to previous interaction with the Advisor. Judges appeared to take advice only if they had developed trust from their previous relationship or their Advisor expressed high confidence. The fact that Judge trust predicted final decision accuracy in experiment one but not two suggests that trust developed from the relationship is less beneficial to the Judge than the trust arising from dependence and expertise asymmetry.

Money. Allocation to Advisor differed significantly between the Advisor-power ($M = 56.85$, $SD = 20.10$) and Judge-power ($M = 41.25$, $SD = 16.14$) conditions, $t(53) = 3.18$, $p < .005$. Those with reward power kept more for themselves in each condition.

GENERAL DISCUSSION

There were several strengths of these experiments. First, participants had to make a decision to receive or give advice in this study, having the option to decline participation. Second, participants were making allocation decisions that involved real money and task decisions with monetary consequences for them. Third, trust was studied as both an attitude and as a behavior. Fourth, the second study used people in ongoing relationships, which is rare in psychology experiments. Fifth, participants were engaging in real social interactions in real time. Finally, the Advisors' confidence assessments were made in a social setting and communicated to another participant.

Uncertainty and Trust

The role in a Judge–Advisor System had important consequences for trust. Across the different settings of the two experiments and the different conditions, Judges developed higher trust than Advisors did, although this did not reach significance in experiment two. Judges had lower levels of relevant expertise and were dependent on their Advisors and thus should have had social uncertainty about the Advisor. Trusting the Advisor was a means to reduce this uncertainty. Alternatively, Judges could have reduced this uncertainty by distancing themselves from the Advisors and relying on themselves (Meyerson, Weick, & Kramer, 1996). However, the Judges' behavior suggests that they developed trust in and depended on their Advisors. They matched the Advisors' recommendations on the majority of the questions. The Judges' trust revealed a willingness to be vulnerable to the Advisors, and their behavior actually put them at risk and vulnerable to any incompetence or malice by the Advisor.

In experiment two, the Advisor had more information about the Judge than in experiment one. This introduced more variance in experiment two and could explain why Judges did not have significantly higher trust levels than Advisors. Advisors could have based their trust in the Judge on factors outside the experimental setting like previous interaction with the Judge or could have had high trust in the Judge despite not being heavily dependent on the Judges during this study.

In both experiments, Judges used more extreme ends of the trust scale than did Advisors, yielding more variability in their trust ratings and showing more discrimination in rating trust in their partner. Because Judges were depending on Advisors for advice and were more vulnerable than the Advisors were, it may have been important to ascertain whether their Advisor could be trusted. Judges' variability in trust ratings suggests that their trust ratings do not stem from a response bias towards high trust. Judges were not using one end of the scale to rate Advisors but were using more extreme ends of the scale than the Advisors. Advisors' lower trust did not reflect low trust. Rather, Advisors usually trusted their partner "about as much as the average person" which was the middle of the scale.

The pretest and use of scores for role assignments may have lowered Judges' self-confidence about computers, making them more likely to rely on and trust another person for advice. Lee and Neville (1994) found that when people have low self-confidence, they are more likely to trust another source for information over themselves. However, the two experiments in this article confound low expertise with the role of decision maker. Therefore, it cannot be determined which is responsible for Judges' higher trust.

Outcomes of Trust

In both experiments, Judge trust was related to the Judge accepting advice and being more confident in their answers based on that advice. This suggests that trust in an Advisor has consequences for a Judges' decisions as well as confidence in those decisions. However, only when the Judge's trust was associated with behaviors of the Advisor during the task (confidence expression and advice elaboration) did it predict decision quality. Although trust may be increased by a prior relationship with an Advisor, trust from these sources may dilute the beneficial effects of trust for quality decisions.

Advisor Cues to Expertise and Trust

Trust is related to Advisor expressed confidence and Advisor's comments to the Judge only in experiment one. In experiment two, trust was related to knowledge of the Advisor's background and previous interaction with the Advisor and not on cues given in the experiment (i.e. expressed confidence or comments).

Mayer et al. (1995) theorize that trust is a "function of the trustee's perceived ability, benevolence, and integrity." In experiment one, cues to Advisor's ability (i.e., high Advisor confidence or Advisor comments) were all the Judge had on which to base trust, and Judge trust was highly correlated with these cues. However, in experiment two, the Judge could have previously formed a judgment of the Advisor to use as a basis of trust. This may have weakened the link between trust and Advisor confidence and comments.

Also, trust was generally higher in experiment two for both the Judges and Advisors. Knowing more about one's partner and having previous interaction with one's partner may increase trust regardless of whether one is giving or receiving advice. Of course, it is possible that prior interaction can provide a reason to distrust one's partner, but there was no evidence of this in experiment two.

Advisor Confidence

In both experiments, Judges were more likely to accept the Advisors' recommendations as Advisor confidence increased. As in prior studies (Buckley & Sniezek, 1990; Sniezek & Buckley, 1995), Advisor's high confidence influenced the Judge. However, the Judge was wise to be influenced by the high confidence given that Advisor confidence had inter- and intravalidity.

With many studies examining confidence, participants make confidence assessments only for the researcher, with no real consequences for the participant. Yet a primary objective of the expression of confidence is to communicate information to another, often a decision maker. Hence, confidence has consequences. In this study, Advisors' confidence assessments were being made for another participant. The validity of these confidence expressions made them a valuable component of the advice with benefits for the Judge's decisions, and in the event of the opportunity for money, both the Judge's and Advisor's outcomes.

Allocation of Money

In experiment one, participants predominantly followed equality norms in allocating money, but in experiment two they gave more money to themselves. This may have been due the confidentiality of the allocation. Schopler, Insko, Drigotas, Wieselquist, Pemberton, and Cox (1995) found that people are more cooperative when their behavior is identifiable than when it is anonymous. In experiment two, one could demand 100% for oneself and the other person would not find out. It is interesting that people had higher trust in experiment two, but were less cooperative in money allocation.

Yet another factor beside secrecy of allocation decisions differed between the two experiments. The first experiment was designed to reflect a situation where people set the price beforehand and price can influence subsequent behavior, as is common when people do not know each other. Here the fee for advice—whether it is offered or requested—can affect trust of the Judge by the Advisor and vice versa. Given their mutual dependence, it was in each partner's self-interest to have the other partner develop trust during the task. By being somewhat more generous than one would otherwise, the Judge could influence the Advisor to give good advice and the Advisor could influence the Judge to make good decisions, thereby improving the chance of getting money. The higher allocations in experiment one may be in part attributable to such strategies. Although the impact of varying allocations could not be examined here, future research could examine it by manipulating the amount allocated to each person or using larger sample sizes to get larger variation in allocations.

The second experiment was designed to capture settings where monetary benefits are an outcome of the exchange—as is often the case in situations like experiment two, in which people know one another. Again, more variation is needed in money allocation to examine it as an outcome variable.

Conclusion

Both experiments had a similar pattern of results. The importance of trust and confidence in the acceptance of advice was confirmed despite the fact that participants interacted in different settings and with differing information about their partner in the two experiments. Four main points stand out from this research. First is the importance of uncertainty and dependence as antecedents for trust. This was highlighted in the independent variable of role. Second is the importance of cues to expertise, like confidence, in building trust—especially when other information about the person is not available. Third, the importance of an ongoing relationship as a basis for trust is highlighted in the relationship between Judge trust and previous interaction with the Advisor in experiment two. Finally, the influence of trust on behavior is seen in the relationship between trust and acceptance of advice.

The difference between the two experiments in the relationship between Judge trust and Advisor confidence and Advisor comments suggests that social interaction should be investigated in further research. Type and amount of social interaction should be included as an independent variable to understand its effects on trust. Trust may develop differently whether one is interacting through writing, voice only, or face-to-face. The number of social cues available may diminish the impact of cues to ability like confidence in the establishment of trust, and different types of social interaction may change the salience of integrity, benevolence, or ability as the basis of trust. As suggested by the lack of relationship between accuracy and trust in experiment two, when there are more social cues available, a person may need to turn up the volume on cues to expertise, like confidence.

Of course, as McGrath and Hollingshead (1993) suggest, the effects of communication medium and type of social interaction may depend on task type. In those rare tasks with a demonstrably correct answer, trust may be based more on ability despite the type of social interaction. However, most real-world tasks do not have an answer that can be known to be correct at the time of the interaction—if ever. For these, integrity and benevolence may be more important to the establishment of trust, especially when people interact face-to-face.

For someone trying to gain a reputation as a trustworthy provider of advice or trying to get people to accept their advice, these results have important implications. Information asymmetry leads to a trust asymmetry and in the absence of other information about a person, confidence may act as an important cue for trust. Dyadic exchanges with an information asymmetry between provider and user of information are ubiquitous. Therefore, understanding the dynamics of how and when trust is established is important for both users and providers of information alike.

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